

# Ali Shafiee

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[Codeforces](#) | [LinkedIn](#) | [IOI Profile](#) | [Google Scholar](#)

*IOI silver medalist and ICPC World Finalist. PhD researcher in probability, stochastic control, and algorithmic game theory.*

## EDUCATION

### Institute of Science and Technology Austria (ISTA)

MSc in Computer Science, GPA 1/5 (highest), advisor *Prof. Krishnendu Chatterjee*

Vienna, Austria

Sep. 2025 – present

### Sharif University of Technology

BSc in Computer Engineering, GPA 3.68/4 (17.03/20)

Tehran, Iran

Sep. 2018 – Aug. 2023

## RESEARCH

Complexity and algorithms for **sequential decision-making under uncertainty**: Markov decision processes with partial observation and with adversarial model uncertainty, plus **algorithmic game theory**. Author lists are alphabetical, per convention in the field.

- [Randomise Alone, Reach as a Team](#). L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026*.
- [Qualitative Analysis of  \$\omega\$ -Regular Objectives on Robust MDPs](#). A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026*.
- [Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete](#). A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025*.
- *Under review*: [On the Complexity of Discounted Robust MDPs with  \$L\_p\$  Uncertainty Sets](#) (NeurIPS 2026); Mechanism Design for Multi-Resource Nakamoto Consensus (*WINE 2026*); Quantitative Analysis of  $\omega$ -Regular Robust MDPs (*AAAI 2027*).

## COMPETITIVE PROGRAMMING

- **Codeforces: top 50 worldwide in 2019**; International Grandmaster, peak rating 2800 (handle ToTLeS)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

## INDUSTRY EXPERIENCE

### Quera, Iran's largest competitive-programming and technical-education platform

Contest Manager → Content Manager → Director of CS Education → Business Line Manager

Tehran, Iran

Mar. 2019 – Aug. 2025

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, Django, Linux/Git, data analysis, machine learning, and deep learning.
- Ran 20+ large-scale programming contests, including sponsored hiring hackathons with corporate partners.

## TEACHING & SERVICE

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, *IOI 2019*, Baku
- **Problem-setter** (2017 – present): Iranian National Olympiad in Informatics, Codecup, and Quera contests
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming
- **Olympiad coach** (2017 – 2023): trained 50+ students for the Iranian National Olympiad in Informatics and the IOI

## SKILLS

**Programming**: C++, Python, Rust · **Numerical & ML**: NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch, JAX

**Tools**: Git, Linux/Bash, SQL, Docker, Jupyter, L<sup>A</sup>T<sub>E</sub>X · **Mathematics**: probability, stochastic processes, Markov decision processes, optimization, game theory

**Languages**: Persian (native), English (TOEFL 110/120)